3/7/26

EXPERIMENT - 2

CISCO PACKET TRACER

AIM:

To study the packet tracer tool installation and user interface.

DESCRIPTION:

This experiment introduced, Cisco packet traces simulation tool and its user interface for designing and simulation computer network. It also covers the basics networking devices such as rules and switch. A hub broadcasts data to all connected devices, whereas a switch forwards data only to the intended destination using mac address, making communication faster and more efficient.

PROCEDURE:

1. Install & launch cisco packet tracer
2. Observe the user interface including device toolbar and workplace
3. Drag and place networking device such as PCs, hubs and switches onto the workspace.

4) Connect the device using the appropriate cables

5) Assign IP address to the PCs

6) Test connectivity between the device using ping command.

7) Observe the data transmission through the hub and the switch

8) Save the network topology and reward observation.

OBSERVATION:

a) Behaviour of switch and hub in forwarding packets

Hub: A hub forwards the received data to all connected device without checking the destination address

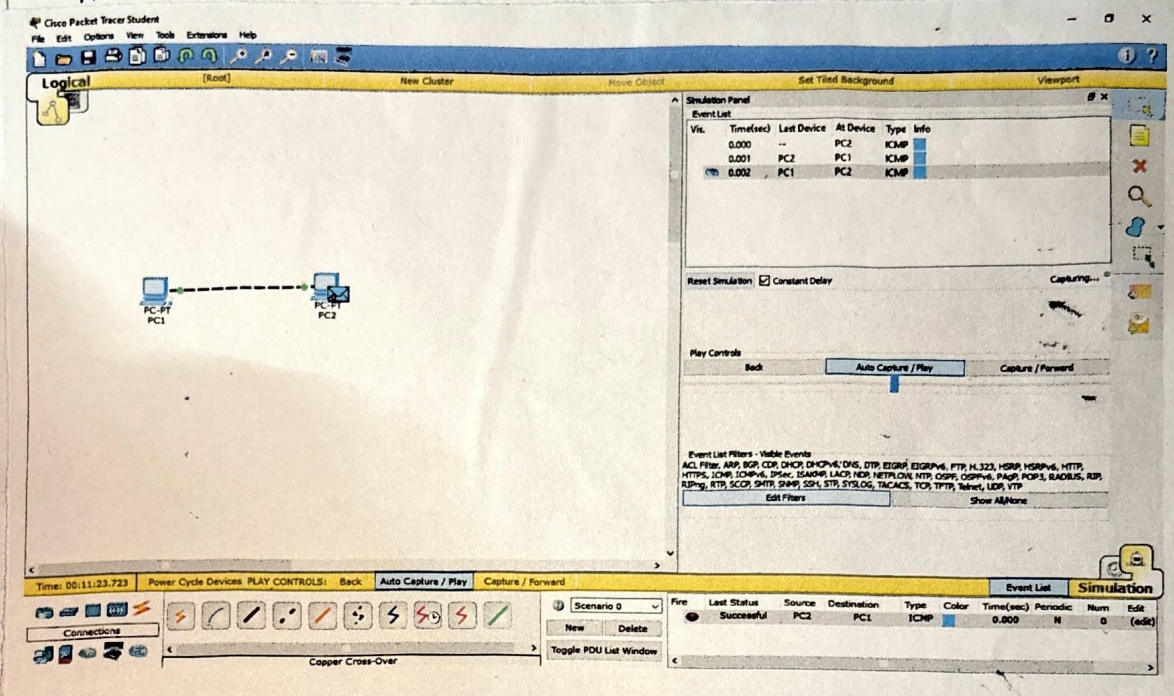
Switch: A switch forwards the received data only to the destination device by using MAC address, making communication faster and more secure

b) Network topology implemented in college

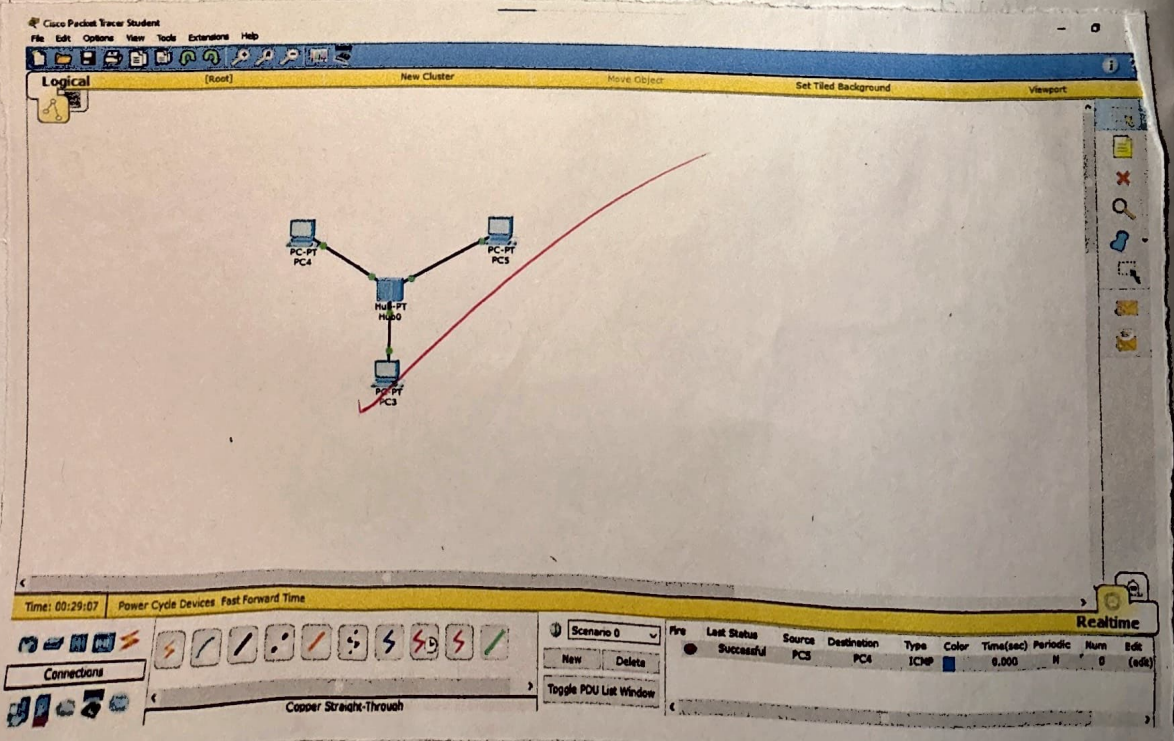
Most colleges use a star topology where all computers are connected in central switch the hub broadcast packet is all connected device

Diagram:

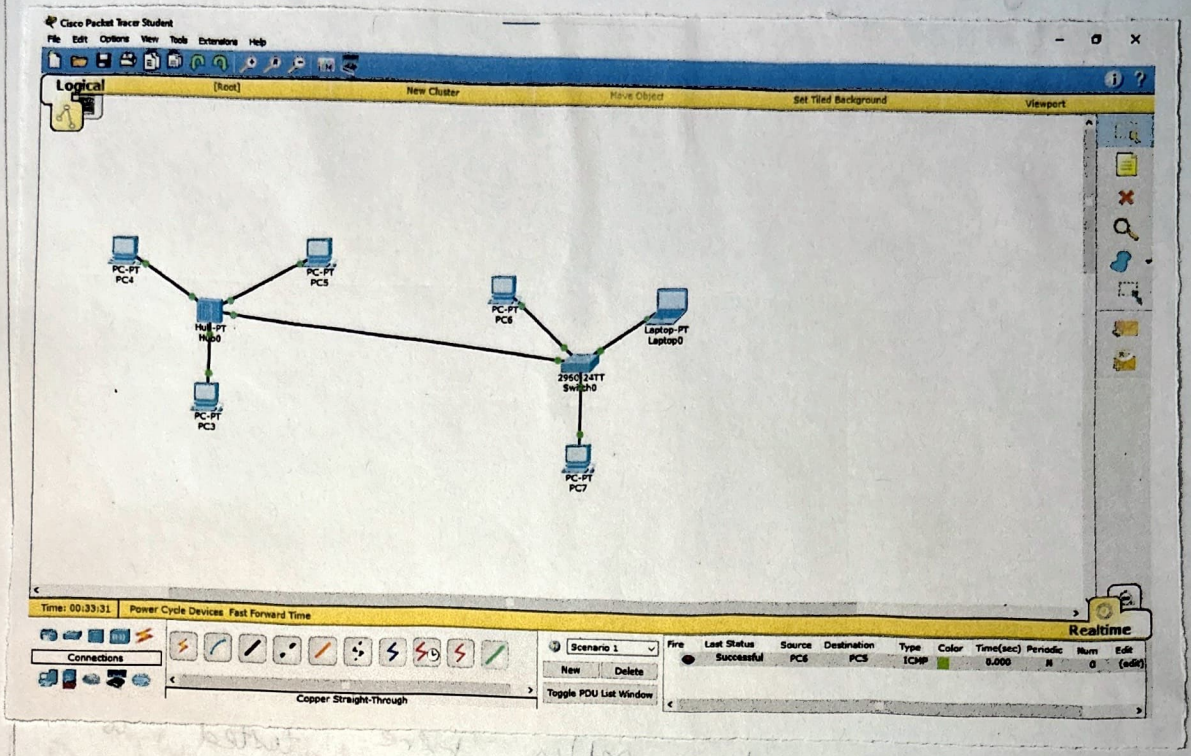
Hub:



Switch:



c. Network :



RESULT:

The Cisco packet tracer environment was successfully explored the working of hubs and switch.